



Electrical Measurment

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➤ INTRODUCTION

- Measurement is the result of a quantitative comparison between a given quantity and a quantity of the same kind chosen as a unit.
- The result of the measurement is expressed by a pointer deflection over a predefined scale or a number representing the ratio between the unknown quantity and the standard.
- The device or instrument used for comparing the unknown quantity with the unit of measurement or a standard quantity is called a measuring instrument.
- The value of the unknown quantity can be measured by direct or indirect methods.

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- In direct measurement methods, the unknown quantity is measured directly instead of comparing it with a standard.
- Examples of direct measurement are current by ammeter, voltage by voltmeter, resistance by ohmmeter, power by wattmeter
- In indirect measurement methods, the value of the unknown quantity is determined by measuring the functionally related quantity and calculating the desired quantity rather than measuring it directly. Suppose the resistance as (R) of a conductor can be measured by measuring the voltage drop across the conductor and dividing the voltage (V) by the current (I) through the conductors, by Ohm's $R = \frac{V}{I}$

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➤ Measurement units

Quantity	Standard unit	Symbol
Length	Meter	m
Mass	kilogram	kg
Time	second	s
Temperature	kelvin	K
Current	ampere	A
Voltage Difference	volt	V
Resistance	ohm	Ω

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➤ Measurement units

Quantity	Standard unit	Symbol
Area	square meter	m^2
Volume	cubic meter	m^3
Velocity	meter per second	m/s
Acceleration	meter per second squared	m/s^2

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➤ Measurement system applications

➤ Present-day applications of measuring instruments can be classified into three major areas.

1. The first of these is their use in regulating trade

- applying instruments that measure physical quantities such as length, volume and mass in terms of standard units.

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➤ Measurement system applications

2. The second application area of measuring instruments is in monitoring functions.

- These provide information that enables human beings to take some prescribed action accordingly.
- The gardener uses a thermometer to determine whether he should turn the heat on in his greenhouse or open the windows if it is too hot.

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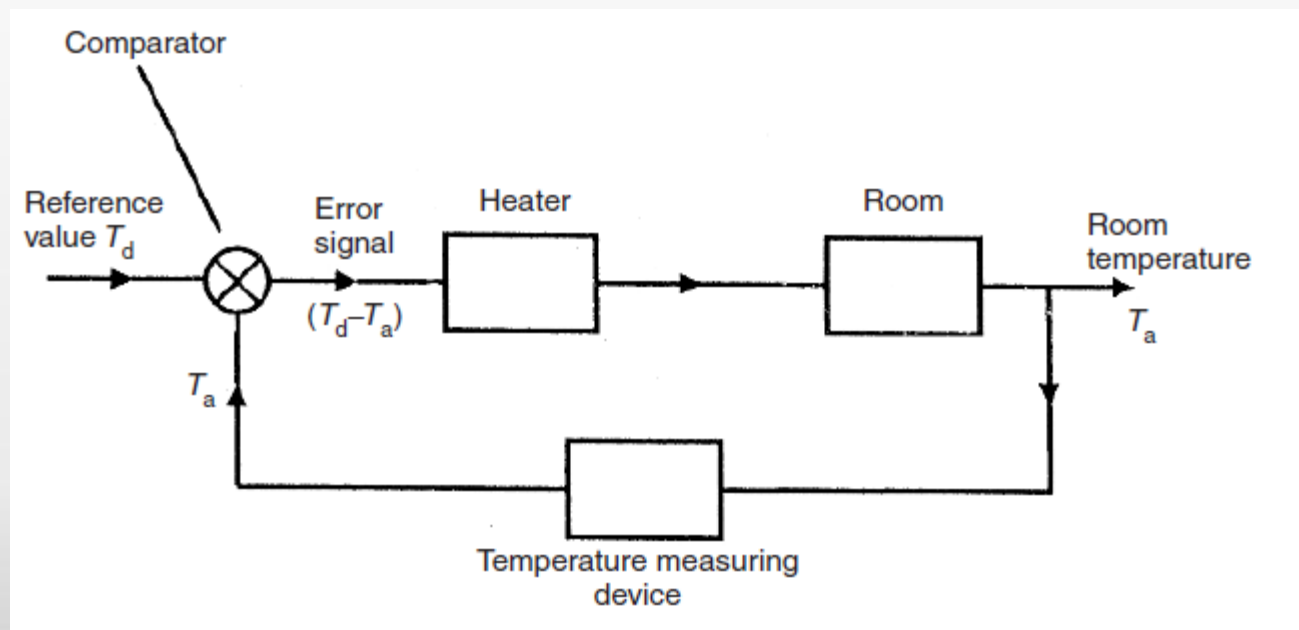
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3. The third application area

- Use as part of automatic feedback control systems forms of measurement systems.



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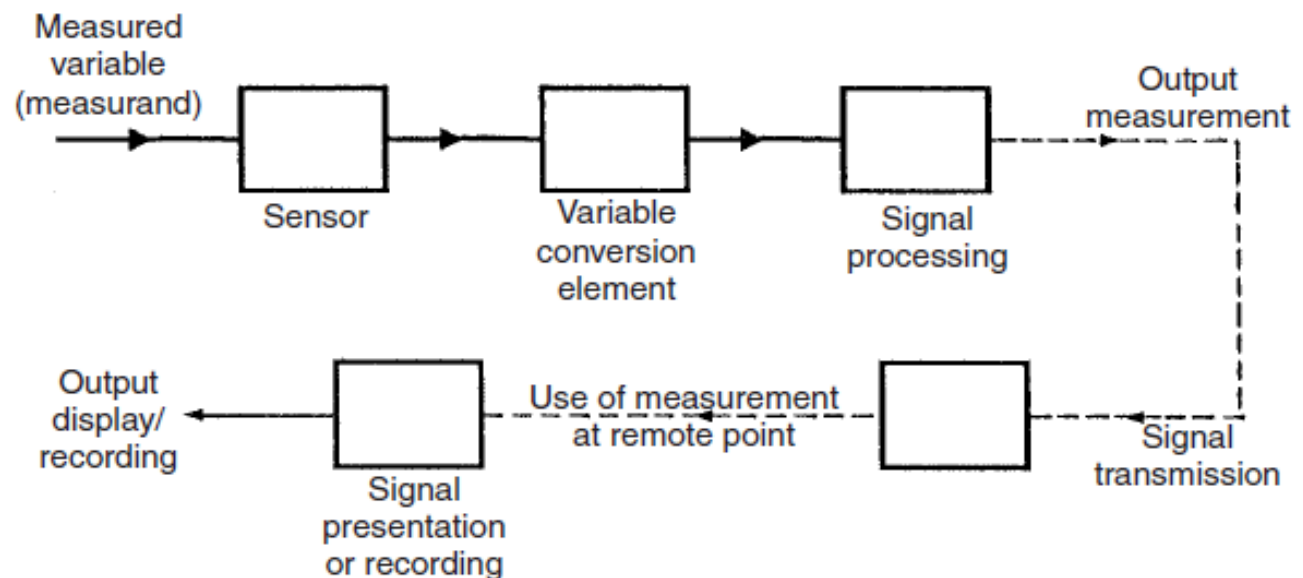
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➤ Elements of a measurement system

1. primary sensor.
2. Variable conversion elements.
3. The Signal processing elements.



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➤ Elements of a measurement system

➤ The first element in any measuring system is the primary sensor:

- this gives an output that is a function of the measurand (the input applied to it).
- For most but not all sensors, this function is at least approximately linear.
- Some examples of primary sensors are a liquid-in-glass thermometer, a thermocouple and a strain gauge.

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➤ Elements of a measurement system

➤ The second element in any measuring system is the Variable conversion elements :

- are needed where the output variable of a primary transducer is in an inconvenient form and has to be converted to a more convenient form.
- For instance, the displacement-measuring strain gauge has an output in the form of a varying resistance. The resistance change cannot be easily measured and so it is converted to a change in voltage by a *bridge circuit*.

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➤ Elements of a measurement system

➤ The second element in any measuring system is the Variable conversion elements :

- In some cases, the primary sensor and variable conversion element are combined, and the combination is known as a *transducer*.

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➤ Elements of a measurement system

➤ The third element in any measuring system is the Signal processing elements :

- exist to improve the quality of the output of a measurement system in some way.
- A very common type of signal processing element is the electronic amplifier, which amplifies the output of the primary transducer or variable conversion element, thus improving the sensitivity and resolution of measurement.

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➤ Elements of a measurement system

➤ In addition to these three components just mentioned, some measurement systems

- have one or two other components, firstly to transmit the signal to some remote point
- and secondly to display or record the signal if it is not fed automatically into a feedback control system.

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➤ Choosing appropriate measuring instruments

- The starting point in choosing the most suitable instrument to use for measurement of a particular quantity in a manufacturing plant or other system is the specification of the instrument characteristics required, especially parameters like the desired measurement accuracy, resolution, sensitivity and dynamic performance.

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➤ Choosing appropriate measuring instruments

- It is also essential to know the environmental conditions that the instrument will be subjected to.
- Cost is very strongly correlated with the performance of an instrument.

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➤ Instrument types

1. Active and passive instruments

- Instruments are divided into active or passive ones according to whether the instrument output is entirely produced by the quantity being measured or whether the quantity being measured simply modulates the magnitude of some external power source.

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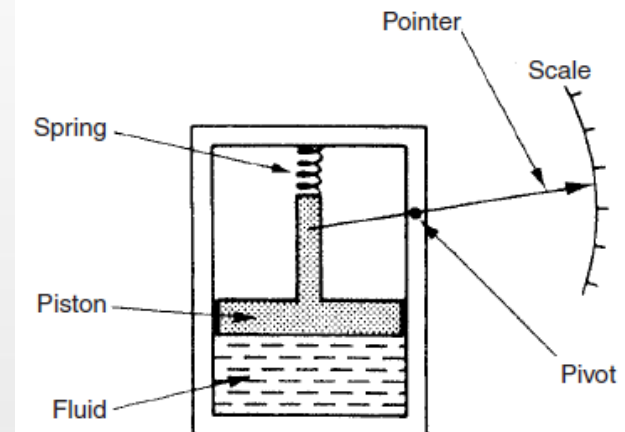
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➤ Instrument types

1. Active and passive instruments

An example of a passive instrument is the pressure-measuring device shown in Figure . The pressure of the fluid is translated into a movement of a pointer against a scale. The energy expended in moving the pointer is derived entirely from the change in pressure measured: there are no other energy inputs to the system.



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➤ Instrument types

1. Active and passive instruments

- it is possible to increase measurement resolution by making the pointer longer, such that the pointer tip moves through a longer arc, the scope for such improvement is clearly restricted by the practical limit of how long the pointer can conveniently be.

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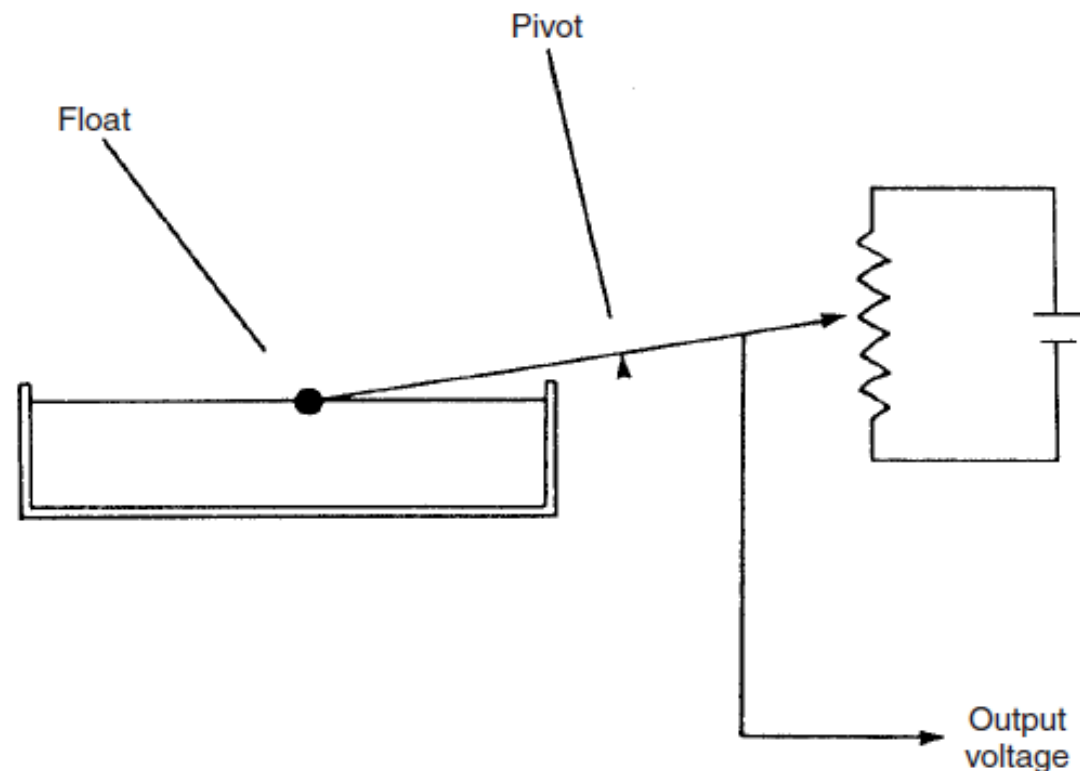
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➤ Instrument types

1. Active and passive instruments



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➤ Instrument types

1. Active and passive instruments

- Adjustment of the magnitude of the external energy input allows much greater control over measurement resolution.

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➤ Instrument types

1. Active and passive instruments

- Whilst the scope for improving measurement resolution is much greater incidentally, it is not infinite because of limitations placed on the magnitude of the external energy input, in consideration of heating effects and for safety reasons.
- In terms of cost, passive instruments are normally of a more simple construction than active ones and are therefore cheaper to manufacture. Therefore, choice between active and passive instruments for a particular application involves carefully balancing the measurement resolution requirements against cost.

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➤ Instrument types

1. Active and passive instruments

- An example of an active instrument is a float-type petrol tank level indicator as sketched in Figure. Here, the change in petrol level moves a potentiometer arm, and the output signal consists of a proportion of the external voltage source applied across the two ends of the potentiometer. The energy in the output signal comes from the external power source: the primary transducer float system is merely modulating the value of the voltage from this external power source.

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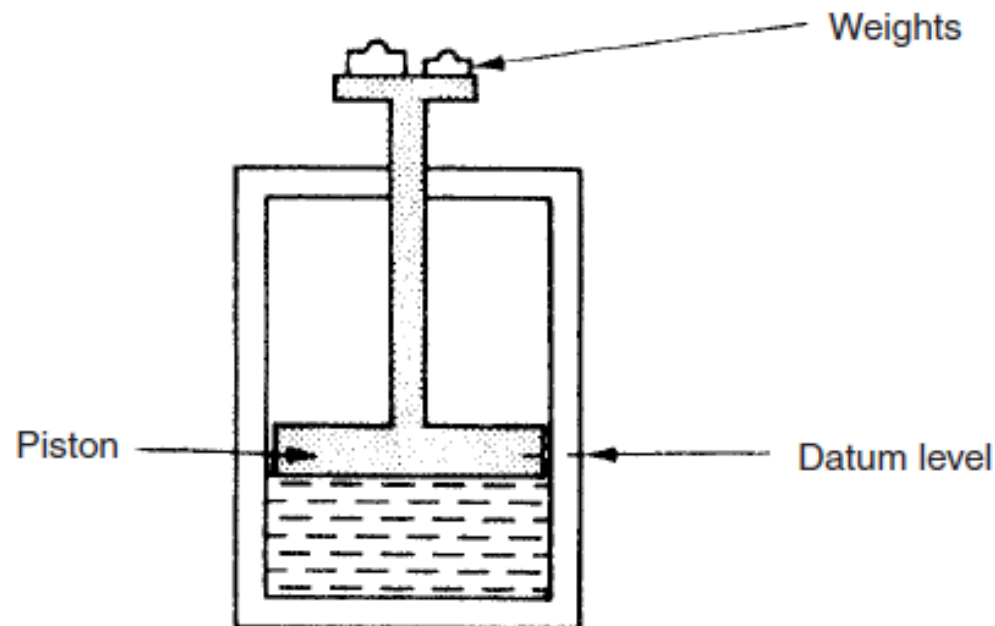
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➤ Instrument types

2. Null-type and deflection-type instruments



terms of the value of the weights needed to reach this null position.

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➤ Instrument types

3. Analogue and digital instruments

- An analogue instrument gives an output that varies continuously as the quantity being measured changes.
- The output can have an infinite number of values within the range that the instrument is designed to measure.

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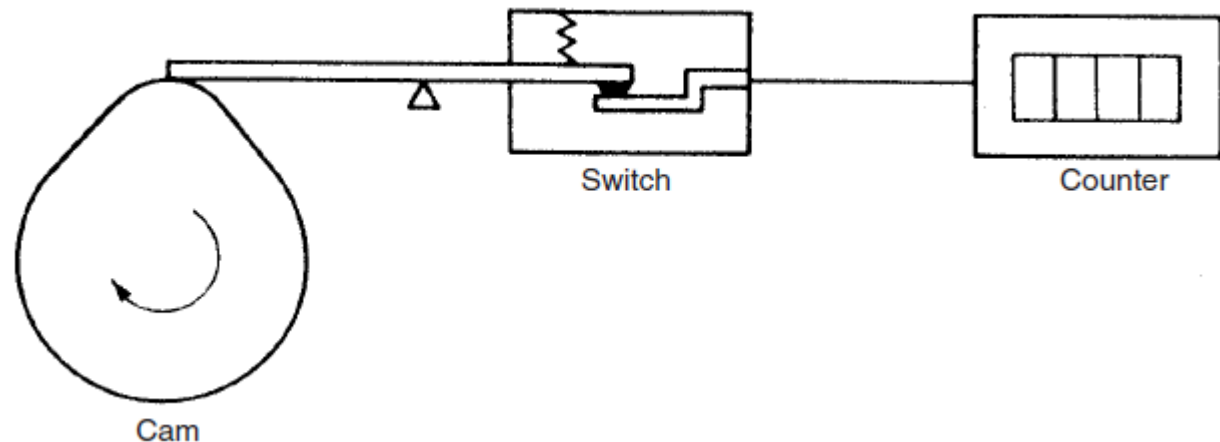
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➤ Instrument types

3. Analogue and digital instruments

- A digital instrument has an output that varies in



a switch. The switching operations are counted by an electronic counter. This system can only count whole revolutions and cannot discriminate any motion that is less than a full revolution.

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3. Analogue and digital instruments

- An instrument whose output is in digital form is therefore particularly advantageous in such applications, as it can be interfaced directly to the control computer. Analogue instruments must be interfaced to the microcomputer by an analogue-to-digital (A/D) converter, which converts the analogue output signal from the instrument into an equivalent digital quantity that can be read into the computer.

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➤ Instrument types

3. Analogue and digital instruments

- This conversion has several disadvantages.
- Firstly, the A/D converter adds a significant cost to the system.
- Secondly, a finite time is involved in the process of converting an analogue signal to a digital quantity, and this time can be critical in the control of fast processes where the accuracy of control depends on the speed of the controlling computer.

Questions and Discussion

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Good Luck
See You in
Next Lecture . . .



With My Best Wishes
Dr. Eng. Mohamed Abdo